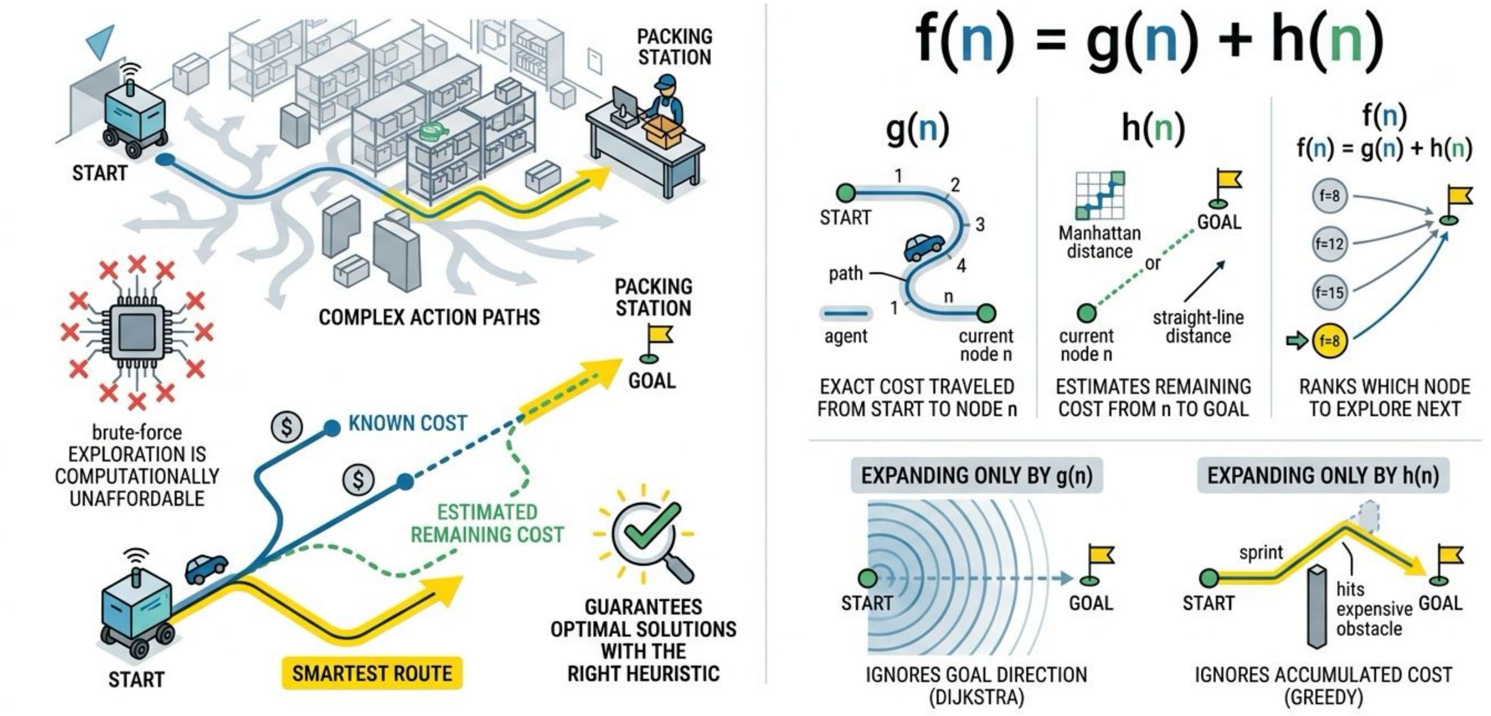
U.S. General Services Administration

Chapter 5.6: A* Search and Replanning

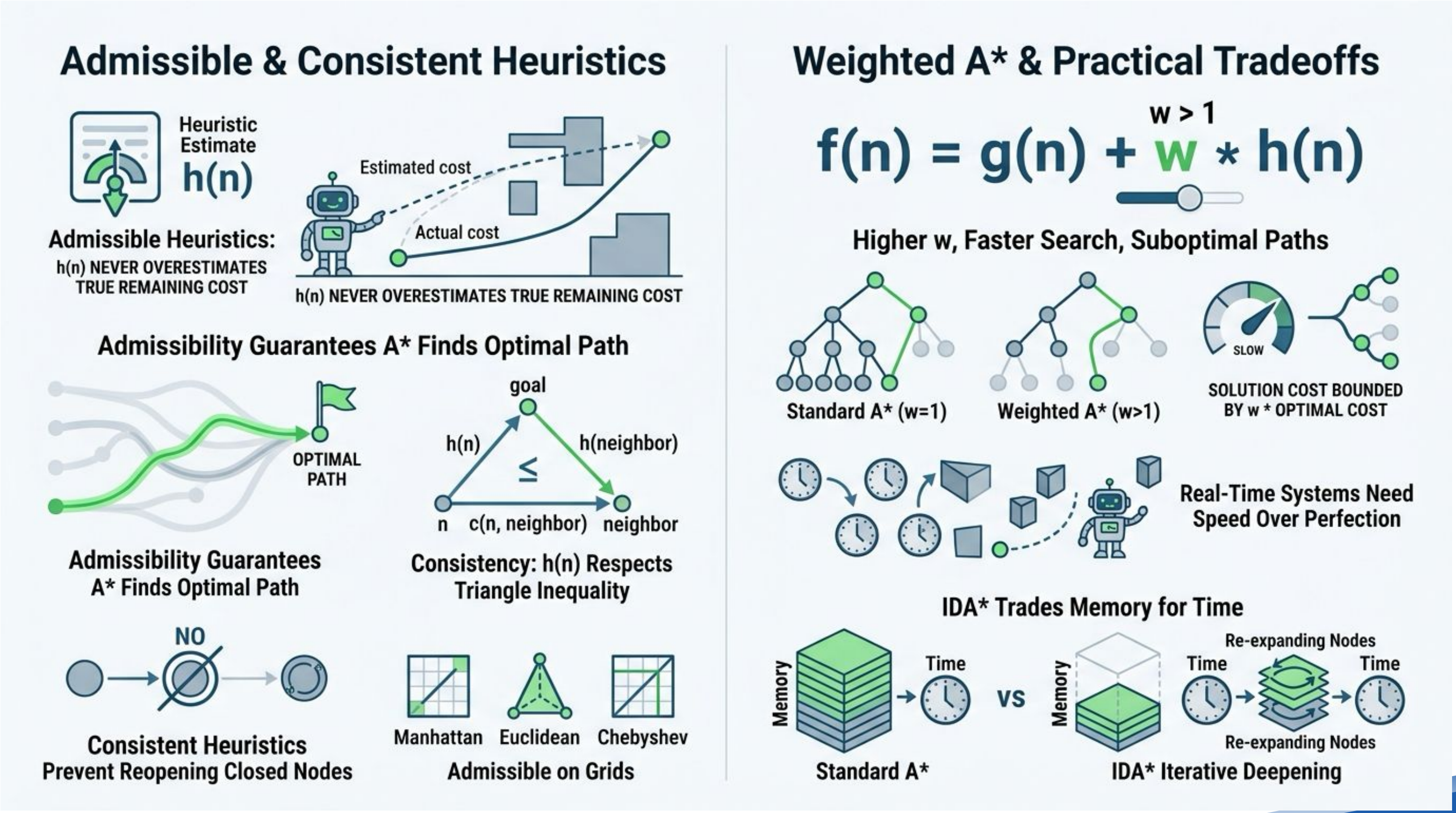
US General Service Administration

AI Community of Practice - March 2026

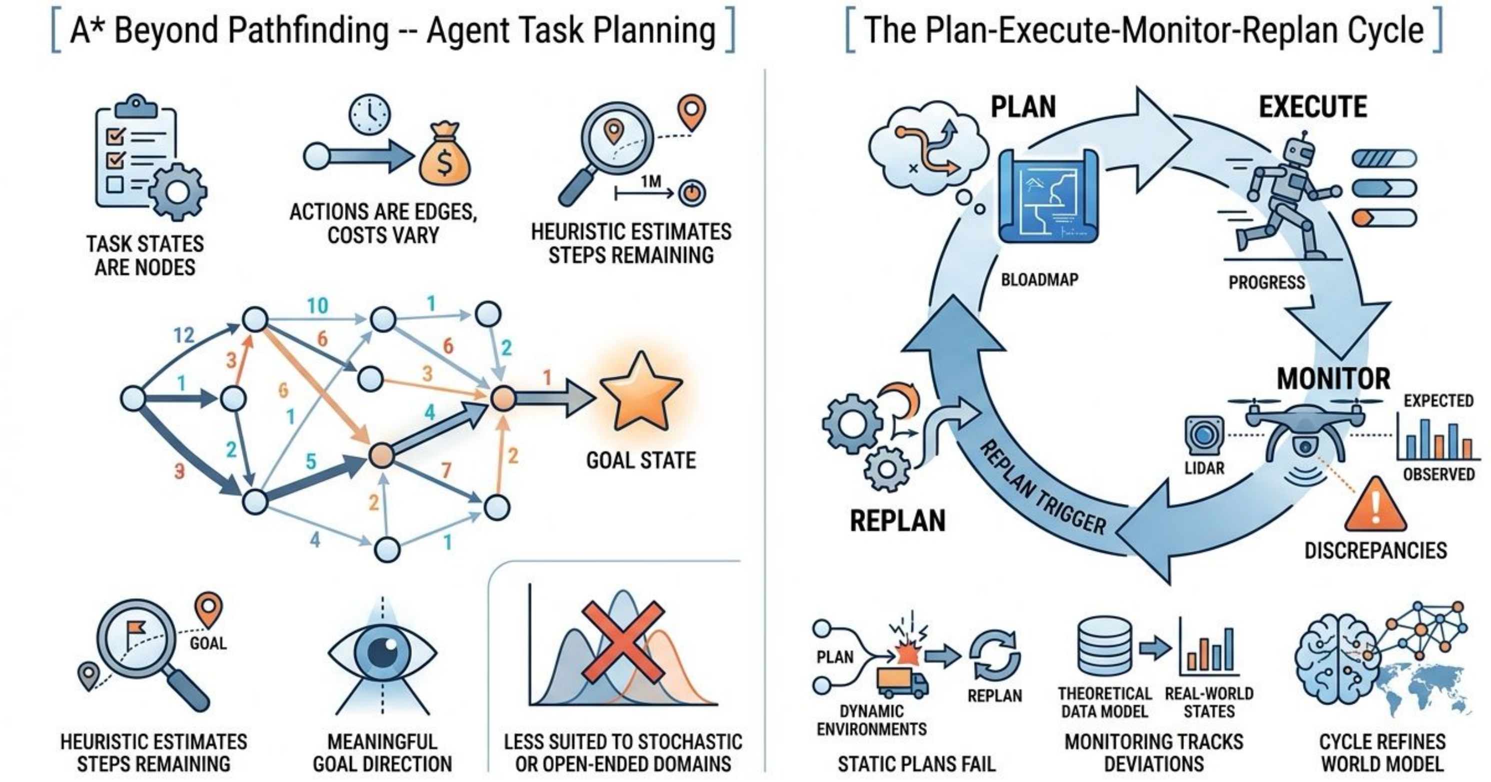
The A* Evaluation Function



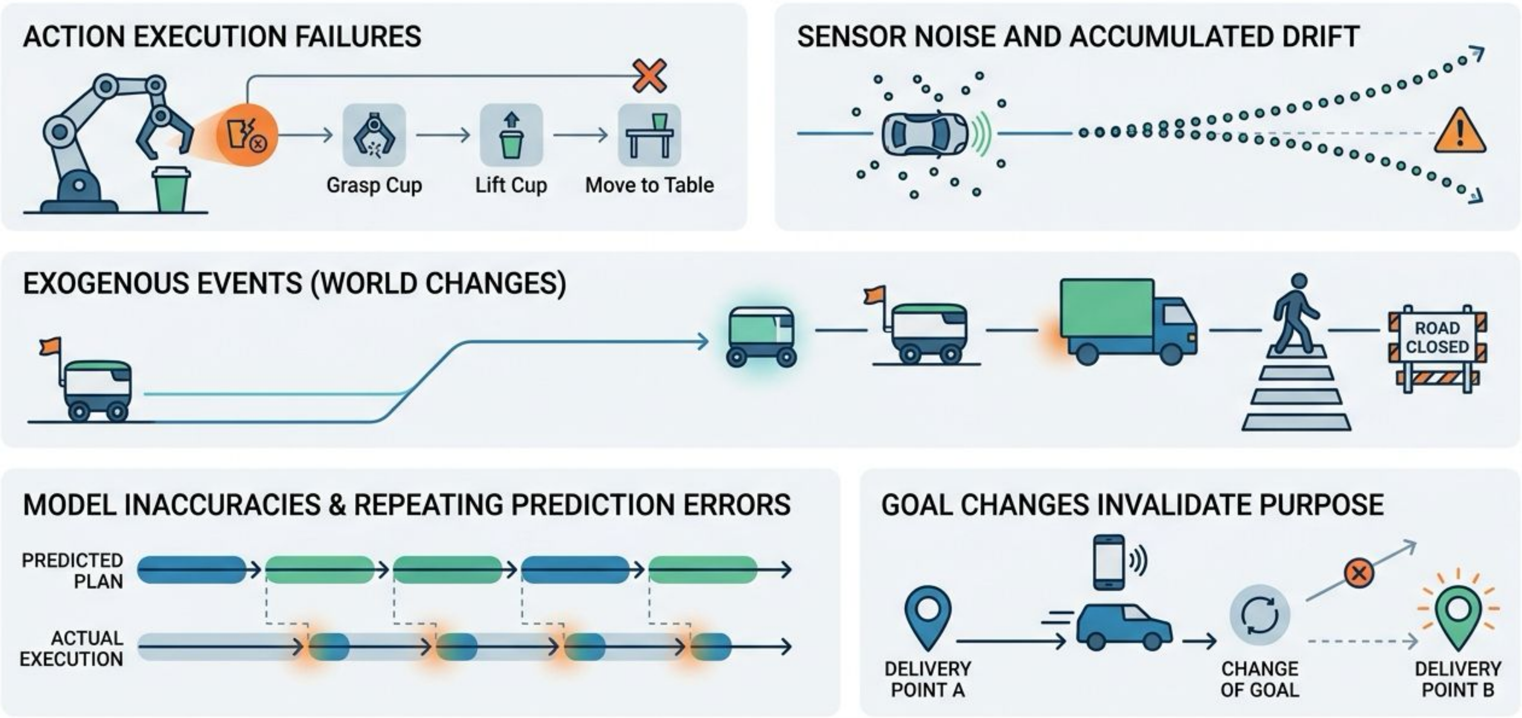
Heuristics & Tradeoffs



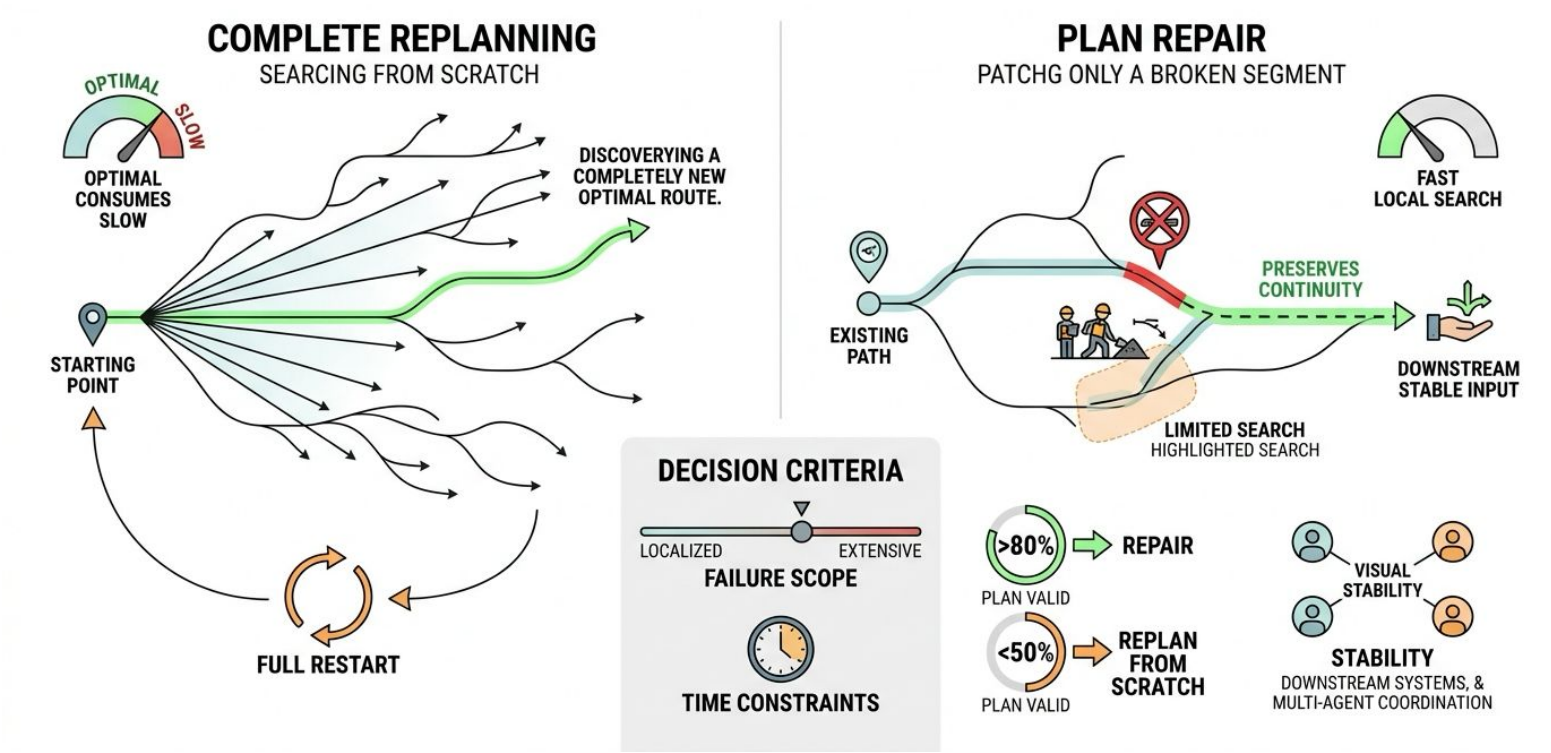
The Plan-Execute-Monitor-Replan Cycle



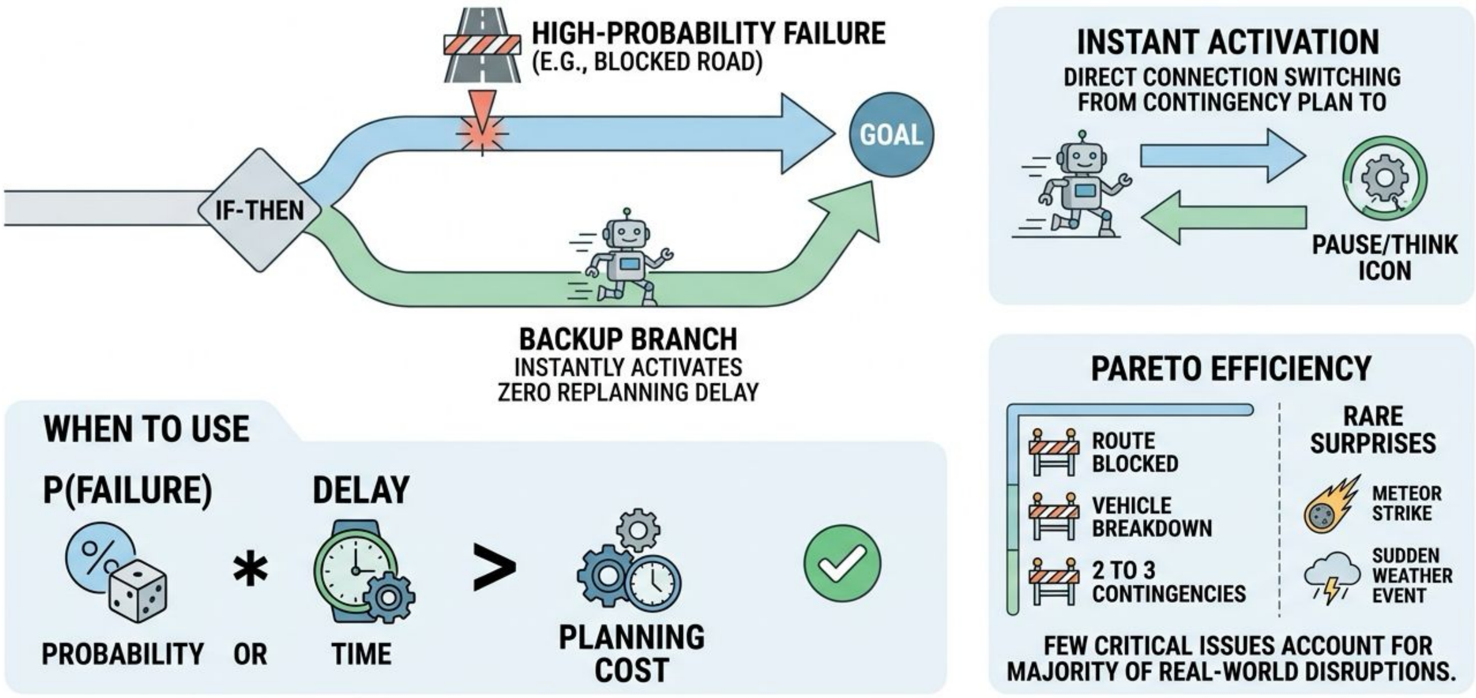
Replanning Triggers



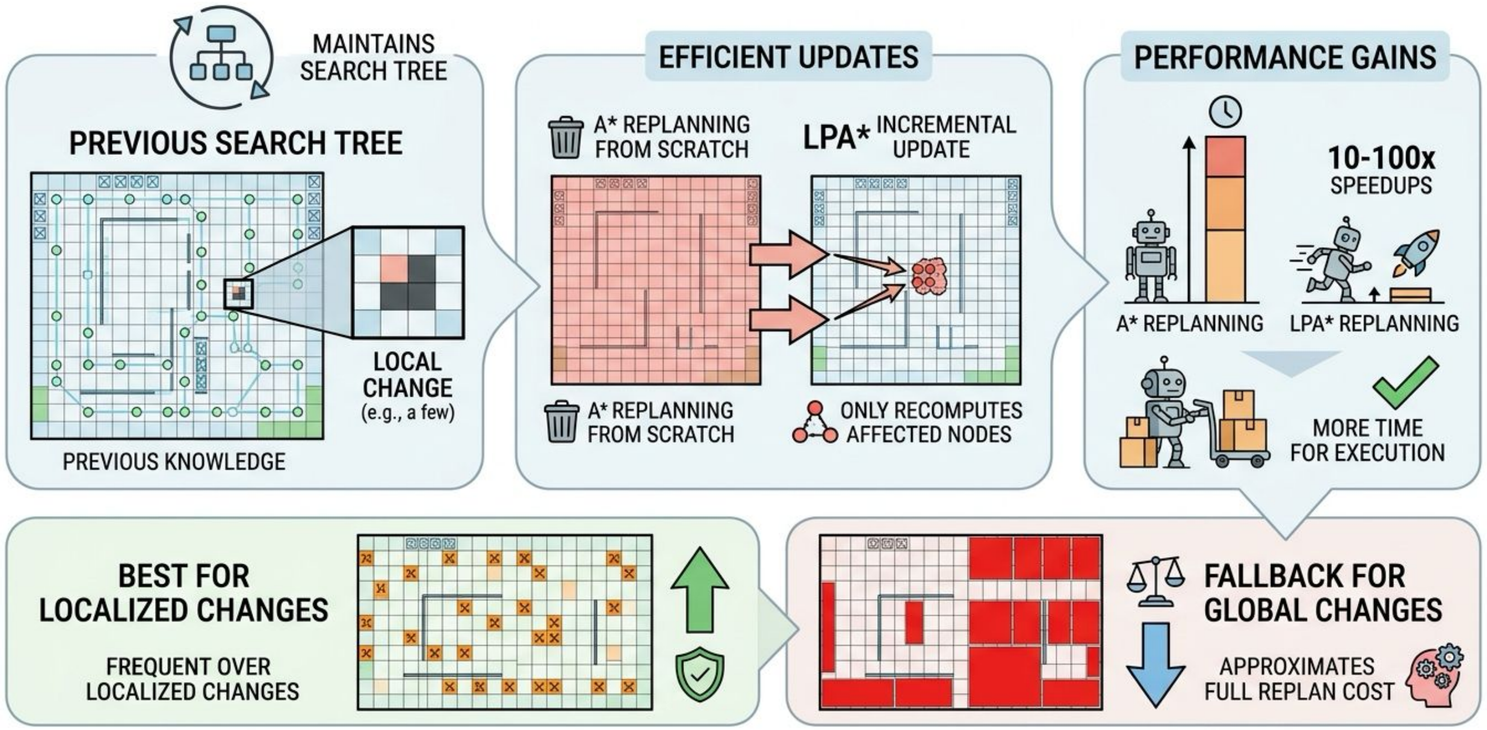
Repair vs. Complete Replanning



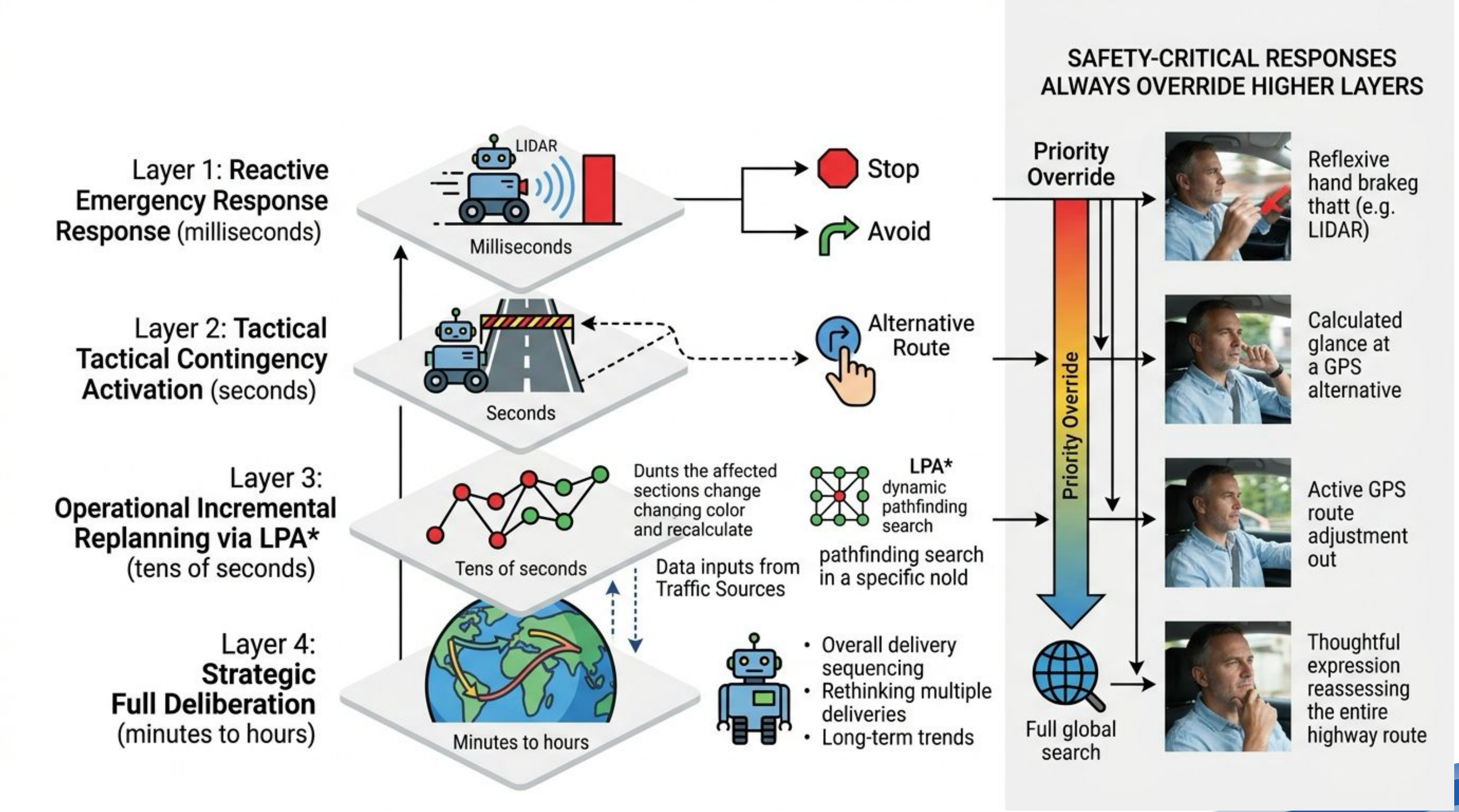
Contingency Planning



LPA* -- Incremental Replanning



Multi-Layered Replanning Architecture



Key Takeaways

- A* balances known cost with heuristic estimation for optimal search
- Admissibility is the mathematical key to optimality guarantees
- Replanning transforms static plans into adaptive behaviors
- Choose repair, contingency, or LPA* based on failure characteristics
- Next chapter: how agents remember past experiences to plan better